

Intro to R - 5. R for Data Science

OIT/SMU Libraries Data Science Workshop Series

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**World Changers
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1 What is Data Science?

2 Predictive Modeling

3 Package Caret

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Section 1

What is Data Science?

Data Science is still evolving. One definition by Hal Varian (Chief economist at Google and professor at UC Berkeley) is:

Data Science

The ability to take data – to be able to understand it, to process it, to extract value from it, to visualize it, to communicate it – that's going to be a hugely important skill in the next decades. – Hal Varian



Figure 1: Data Science LifeCycle

Source: <https://datascience.berkeley.edu/about/what-is-data-science/>

Section 2

Predictive Modeling

- Data mining
- Machine learning
- Prediction
 - ▶ regression (predict a number, e.g., the age of a person)
 - ▶ classification (predict a label, e.g., yes/no)

Predictive Modeling Workflow

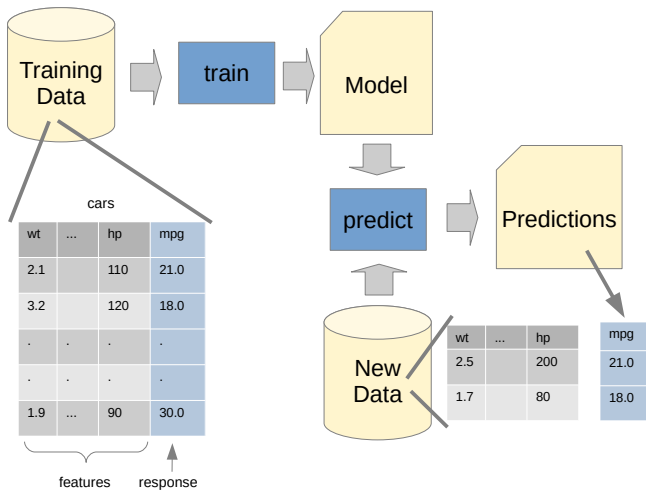


Figure 2: Workflow of Predictive Modeling

Predictive Modeling Workflow in R

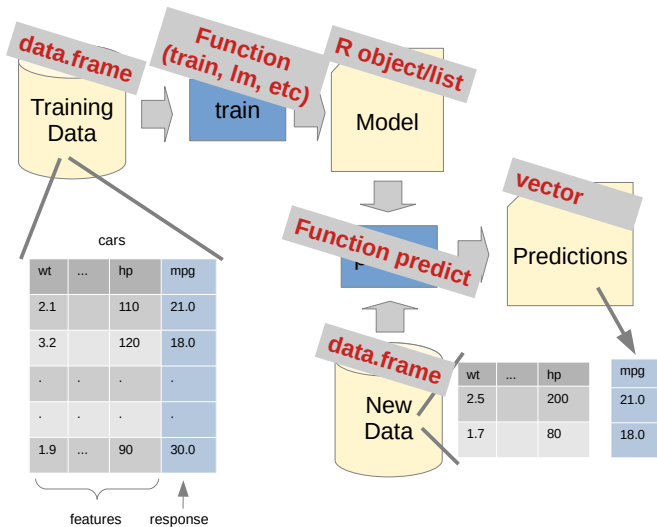


Figure 3: Workflow of Predictive Modeling with R

Example

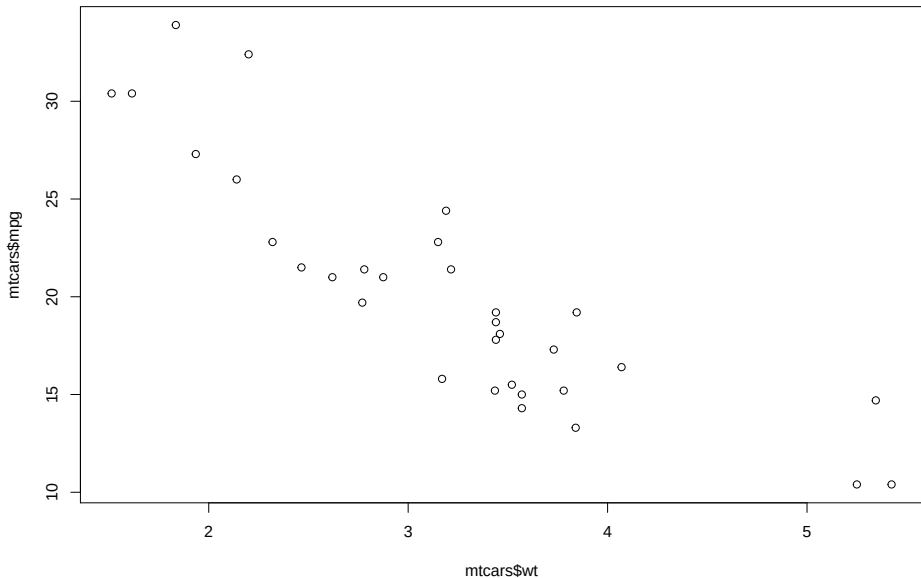
```
data(mtcars)      # Load the dataset
knitr::kable(head(mtcars))
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21	6	160	110	3.9	2.6	16	0	1	4	4
Mazda RX4 Wag	21	6	160	110	3.9	2.9	17	0	1	4	4
Datsun 710	23	4	108	93	3.9	2.3	19	1	1	4	1
Hornet 4 Drive	21	6	258	110	3.1	3.2	19	1	0	3	1
Hornet Sportabout	19	8	360	175	3.1	3.4	17	0	0	3	2
Valiant	18	6	225	105	2.8	3.5	20	1	0	3	1

Note: `kable` in package `knitr` is used to pretty-print the table because the slides were created with Markdown.

Example: Predict Miles per Gallon

```
plot(mtcars$wt, mtcars$mpg)
```



Linear Regression

```
model <- lm(mpg ~ wt, data = mtcars)
model

##
## Call:
## lm(formula = mpg ~ wt, data = mtcars)
##
## Coefficients:
## (Intercept)          wt
##      37.29      -5.34
```

Formula Interface

R often uses a “model formula” to specify models of the form response ~ predictors. See ? formula for details.

Linear Regression: Model summary

```
summary(model)
```

```
##
## Call:
## lm(formula = mpg ~ wt, data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.543 -2.365 -0.125  1.410  6.873
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   37.285     1.878   19.86 < 2e-16 ***
## wt           -5.344     0.559   -9.56 1.3e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3 on 30 degrees of freedom
## Multiple R-squared:  0.753, Adjusted R-squared:  0.745
## F-statistic: 91.4 on 1 and 30 DF,  p-value: 1.29e-10
```

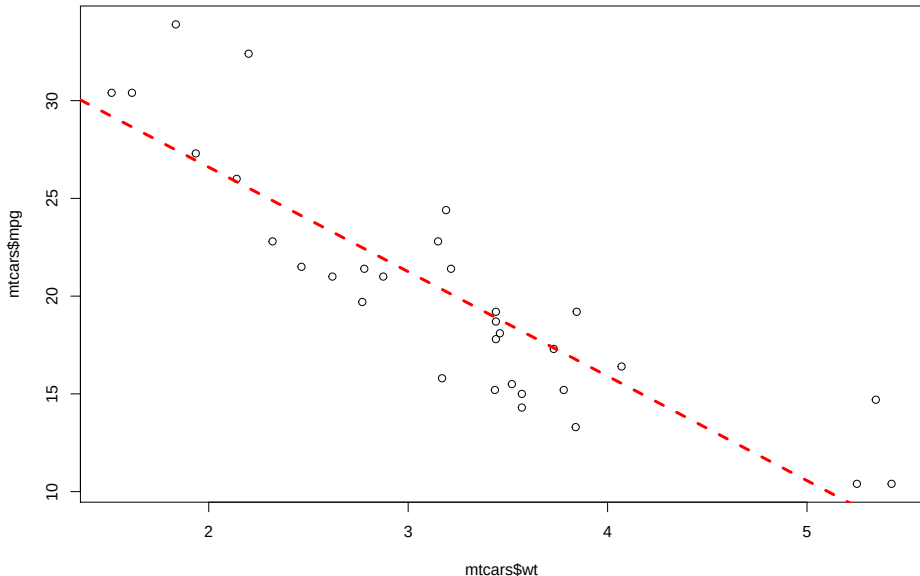
Linear Regression: The model as an R object

```
str(model)
```

```
## List of 12
## $ coefficients : Named num [1:2] 37.29 -5.34
## ..- attr(*, "names")= chr [1:2] "(Intercept)" "wt"
## $ residuals    : Named num [1:32] -2.28 -0.92 -2.09 1.3 -0.2 ...
## ..- attr(*, "names")= chr [1:32] "Mazda RX4" "Mazda RX4 Wag" "Datsun 7
## $ effects      : Named num [1:32] -113.65 -29.116 -1.661 1.631 0.111 ..
## ..- attr(*, "names")= chr [1:32] "(Intercept)" "wt" "" "" ...
## $ rank         : int 2
## $ fitted.values: Named num [1:32] 23.3 21.9 24.9 20.1 18.9 ...
## ..- attr(*, "names")= chr [1:32] "Mazda RX4" "Mazda RX4 Wag" "Datsun 7
## $ assign       : int [1:2] 0 1
## $ qr           :List of 5
## ..$ qr        : num [1:32, 1:2] -5.657 0.177 0.177 0.177 0.177 ...
## .. ..- attr(*, "dimnames")=List of 2
## .. .. ..$ : chr [1:32] "Mazda RX4" "Mazda RX4 Wag" "Datsun 710" "Horne
## .. .. ..$ : chr [1:2] "(Intercept)" "wt"
## .. ..- attr(*, "assign")= int [1:2] 0 1
## ..$ qraux: num [1:2] 1.18 1.05
## ..$ pivot: int [1:2] 1 2
## ..$ tol   : num 1e-07
## ..$ cond_
```

Linear Regression: Plotting the regression line

```
plot(mtcars$wt, mtcars$mpg)
abline(coef(model), col = "red", lty = 2, lwd = 3)
```



Multiple Linear Regression

```
model <- lm(mpg ~ wt + cyl + hp, data = mtcars)
model
```

```
##
## Call:
## lm(formula = mpg ~ wt + cyl + hp, data = mtcars)
##
## Coefficients:
## (Intercept)          wt          cyl          hp
##      38.752      -3.167      -0.942      -0.018
```

```
summary(model)
```

```
##
## Call:
## lm(formula = mpg ~ wt + cyl + hp, data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.929 -1.560 -0.531  1.185  5.899
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  38.7518     1.7869   21.69  <2e-16 ***
## wt          -3.1670     0.7406   -4.28  0.0002 ***
## cyl         -0.9416     0.5509   -1.71  0.0985 .
## hp          -0.0180     0.0119   -1.52  0.1400
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.5 on 28 degrees of freedom
## Multiple R-squared:  0.843, Adjusted R-squared:  0.826
```

Prediction

Almost all R models provide a `predict` function.

```
predict(model, head(mtcars))
```

##	Mazda RX4	Mazda RX4 Wag	Datsun 710	Hornet 4 Drive
##	23	22	26	21
##	Hornet Sportabout	Valiant		
##	17	20		

Note: Prediction is typically done on new or test data. Package like `caret` and `mlr3` and `Superlearner` help with this.

Section 3

Package Caret

Train a model with Caret

```
library("caret")

## Loading required package: lattice
## Loading required package: ggplot2

# Simple linear regression model (lm means linear model)
model <- train(mpg ~ wt + cyl + hp,
               data = mtcars,
               method = "lm")

model

## Linear Regression
##
## 32 samples
## 3 predictor
##
## No pre-processing
## Resampling: Bootstrapped (25 reps)
## Summary of sample sizes: 32, 32, 32, 32, 32, 32, ...
## Resampling results:
##
##   RMSE   Rsquared   MAE
##   2.8    0.84      2.3
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

Training a regression tree

```
# rpart implements CART (here a regression tree)
model <- train(mpg ~ wt + cyl + hp,
               data = mtcars,
               method = "rpart")

## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info =
## trainInfo, : There were missing values in resampled performance measures.

model

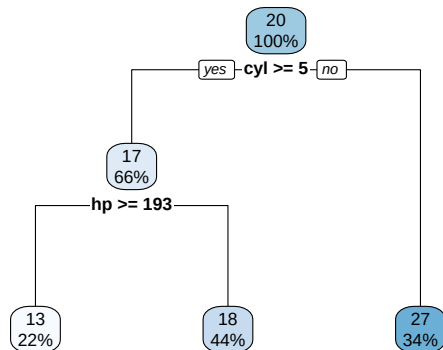
## CART
##
## 32 samples
## 3 predictor
##
## No pre-processing
## Resampling: Bootstrapped (25 reps)
## Summary of sample sizes: 32, 32, 32, 32, 32, 32, ...
## Resampling results across tuning parameters:
##
##    cp      RMSE  Rsquared  MAE
##  0.000   4.0    0.55      3.3
##  0.097   4.1    0.54      3.4
##  0.643   5.1    0.48      4.2
##
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was cp = 0.
```

Plotting a regression tree

```
library(rpart.plot)
```

```
## Loading required package: rpart
```

```
rpart.plot(model$finalModel)
```



```
varImp(model)
```

```
## rpart variable importance
```

```
##
```

```
## Overall
```

```
## hp 100.0
```

```
## cyl 94.6
```

```
## wt 0.0
```

Section 4

Exercises

- 1 Load the MLB data set and create a scatter plot of weight by height with a regression line added.
- 2 Create a prediction model that predicts weight using position, height, and age of the player. Compare different models using caret.
- 3 Create a classification model to predict the position. Decide what information is useful for this task.